

ASSIGNMENT NO. 11: Indexing and Query Optimization Queries

Questions

Q1) Create a MySQL database named SalesDB and a table named OrderData.

OrderData Table

Column Name	Data Type	Constraint
OrderID	INT	PRIMARY KEY
CustomerName	VARCHAR(50)	NOT NULL
Email	VARCHAR(80)	NOT NULL
City	VARCHAR(30)	NOT NULL
ProductCategory	VARCHAR(30)	NOT NULL
OrderDate	DATE	NOT NULL
OrderAmount	DECIMAL(10,2)	NOT NULL
OrderStatus	VARCHAR(20)	NOT NULL

Insert at least 12 records into the table.

Answer:-

```
CREATE TABLE OrderData (  
    OrderID INT PRIMARY KEY,  
    CustomerName VARCHAR(50) NOT NULL,  
    Email VARCHAR(80) NOT NULL,  
    City VARCHAR(30) NOT NULL,  
    ProductCategory VARCHAR(30) NOT NULL,  
    OrderDate DATE NOT NULL,  
    OrderAmount DECIMAL(10,2) NOT NULL,  
    OrderStatus VARCHAR(20) NOT NULL
```

);

The screenshot shows the phpMyAdmin interface with the 'salesdb' database selected. The 'SQL' tab is active, displaying a successful query execution result: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0006 seconds.)'. The query executed was:

```
CREATE TABLE OrderData ( OrderID INT PRIMARY KEY, CustomerName VARCHAR(50) NOT NULL, Email VARCHAR(80) NOT NULL, City VARCHAR(30) NOT NULL, ProductCategory VARCHAR(30) NOT NULL, OrderDate DATE NOT NULL, OrderAmount DECIMAL(10,2) NOT NULL, OrderStatus VARCHAR(20) NOT NULL );
```

The left sidebar shows the database structure, including 'salesdb' and its sub-databases like 'orderdata'.

The screenshot shows the phpMyAdmin interface with the 'salesdb' database selected. The 'SQL' tab is active, displaying a successful query execution result: '12 rows inserted. (Query took 0.0038 seconds.)'. The query executed was:

```
INSERT INTO OrderData ( OrderID, CustomerName, Email, City, ProductCategory, OrderDate, OrderAmount, OrderStatus ) VALUES (1, 'Rahul Sharma', 'rahul@gmail.com', 'Mumbai', 'Electronics', '2026-01-05', 55000.00, 'Delivered'), (2, 'Priya Patil', 'priya@gmail.com', 'Pune', 'Furniture', '2026-01-08', 32000.00, 'Delivered'), (3, 'Aman Verma', 'aman@gmail.com', 'Nashik', 'Clothing', '2026-01-12', 8500.00, 'Pending'), (4, 'Sneha Joshi', 'sneha@gmail.com', 'Mumbai', 'Electronics', '2026-01-20', 72000.00, 'Delivered'), (5, 'Karan Mehta', 'karan@gmail.com', 'Thane', 'Furniture', '2026-02-03', 28000.00, 'Cancelled'), (6, 'Neha Shah', 'neha@gmail.com', 'Pune', 'Electronics', '2026-02-08', 48000.00, 'Delivered'), (7, 'Rohan Desai', 'rohan@gmail.com', 'Mumbai', 'Clothing', '2026-02-15', 12000.00, 'Pending');
```

The left sidebar shows the database structure, including 'salesdb' and its sub-databases like 'orderdata'.

The screenshot shows the phpMyAdmin interface with the 'salesdb' database selected. The 'Table: OrderData' is displayed, showing 12 rows of data. The table structure is as follows:

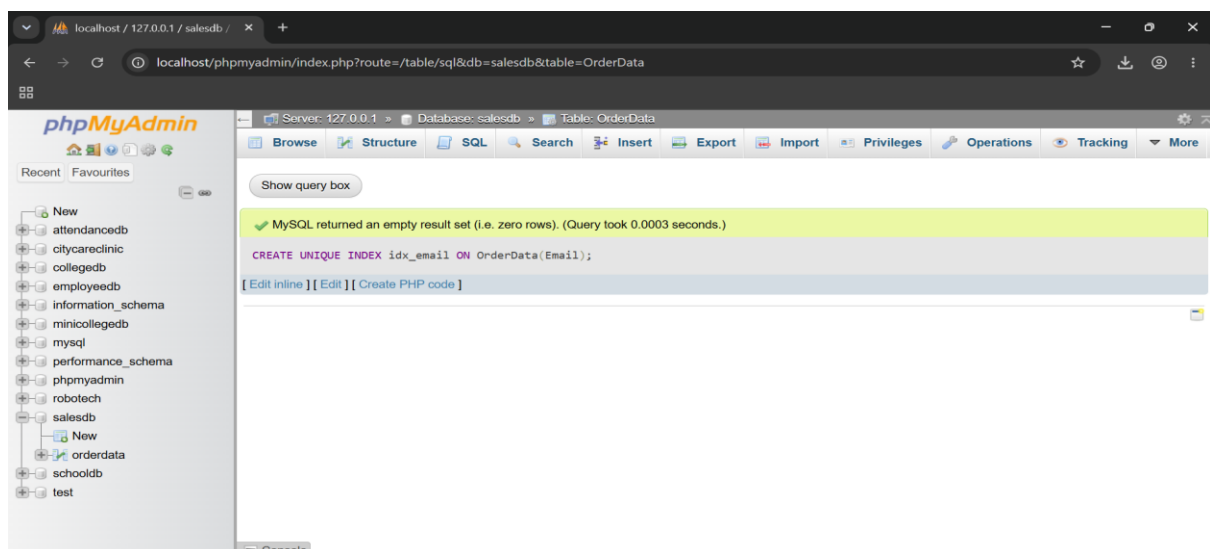
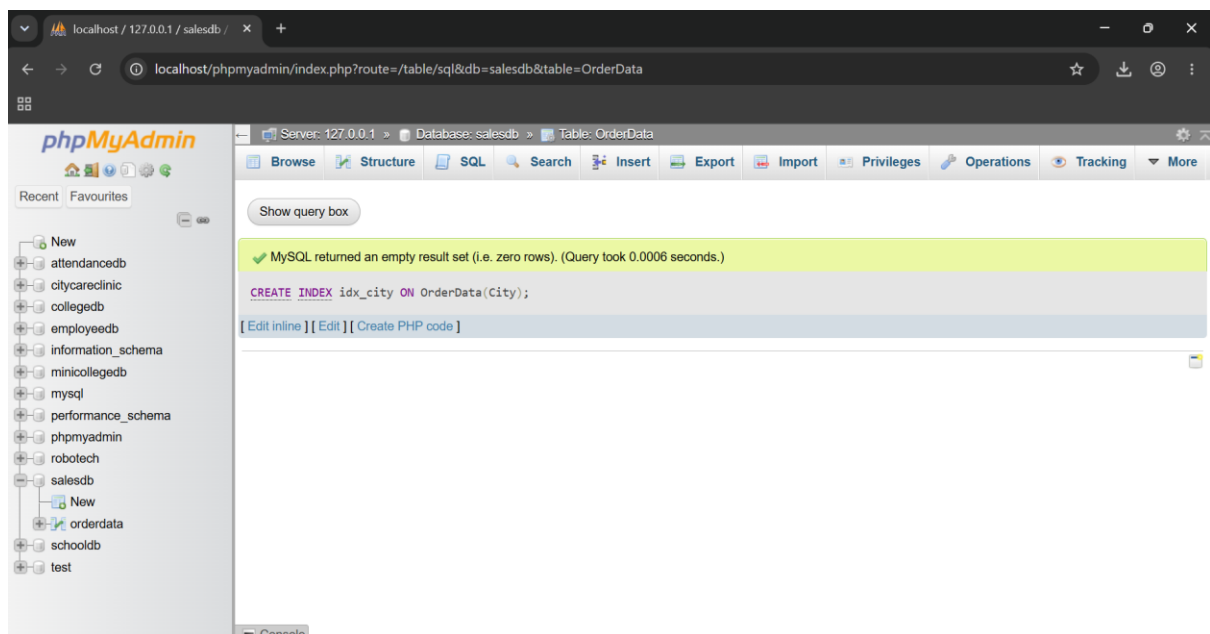
OrderID	CustomerName	Email	City	ProductCategory	OrderDate	OrderAmount	OrderStatus
1	Rahul Sharma	rahul@gmail.com	Mumbai	Electronics	2026-01-05	55000.00	Delivered
2	Priya Patil	priya@gmail.com	Pune	Furniture	2026-01-08	32000.00	Delivered
3	Aman Verma	aman@gmail.com	Nashik	Clothing	2026-01-12	8500.00	Pending
4	Sneha Joshi	sneha@gmail.com	Mumbai	Electronics	2026-01-20	72000.00	Delivered
5	Karan Mehta	karan@gmail.com	Thane	Furniture	2026-02-03	28000.00	Cancelled
6	Neha Shah	neha@gmail.com	Pune	Electronics	2026-02-08	48000.00	Delivered
7	Rohan Desai	rohan@gmail.com	Mumbai	Clothing	2026-02-15	12000.00	Pending
8	Meera Kulkarni	meera@gmail.com	Nashik	Furniture	2026-02-22	36000.00	Delivered
9	Arjun Nair	arjun@gmail.com	Thane	Electronics	2026-03-02	64000.00	Shipped
10	Divya More	divya@gmail.com	Mumbai	Furniture	2026-03-07	41000.00	Delivered
11	Isha Singh	isha@gmail.com	Pune	Clothing	2026-03-12	15000.00	Shipped
12	Om Jadhav	om@gmail.com	Mumbai	Electronics	2026-03-18	59000.00	Pending

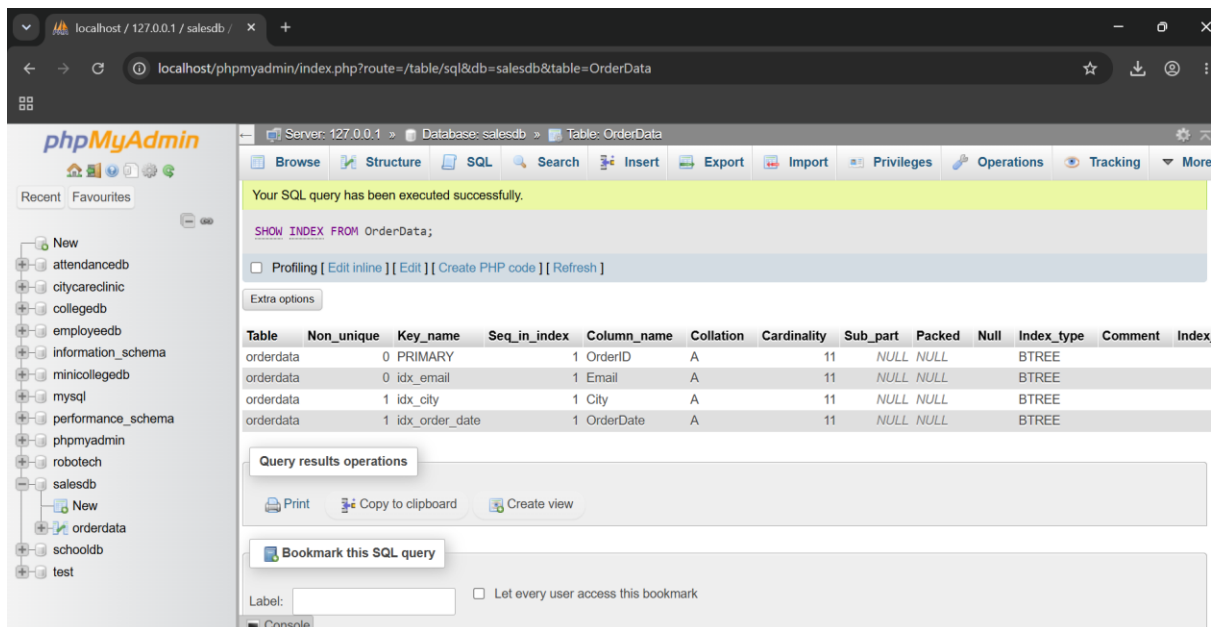
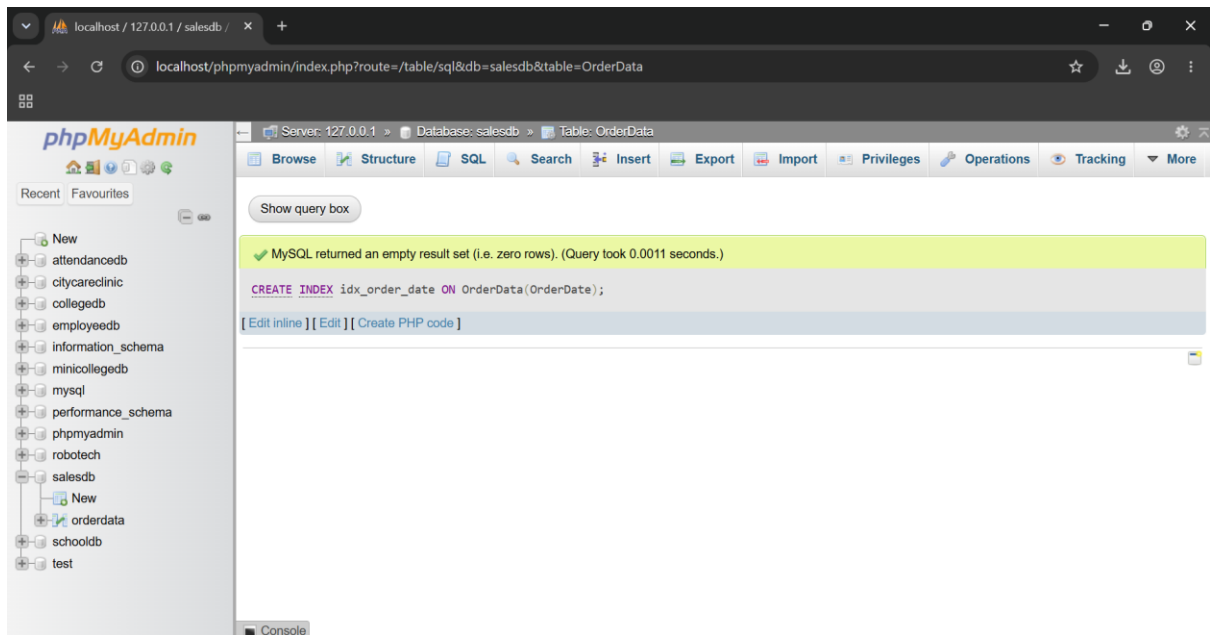
Q2) Create the following MySQL indexes on the OrderData table:

Index Name	Column	Index Type
idx_city	City	Normal index
idx_email	Email	Unique index
idx_order_date	OrderDate	Normal index

Display all indexes using SHOW INDEX.

Answer:-





Q3) Create a composite index named idx_city_order_date.

The index should contain:

Index Position	Column
1	City
2	OrderDate

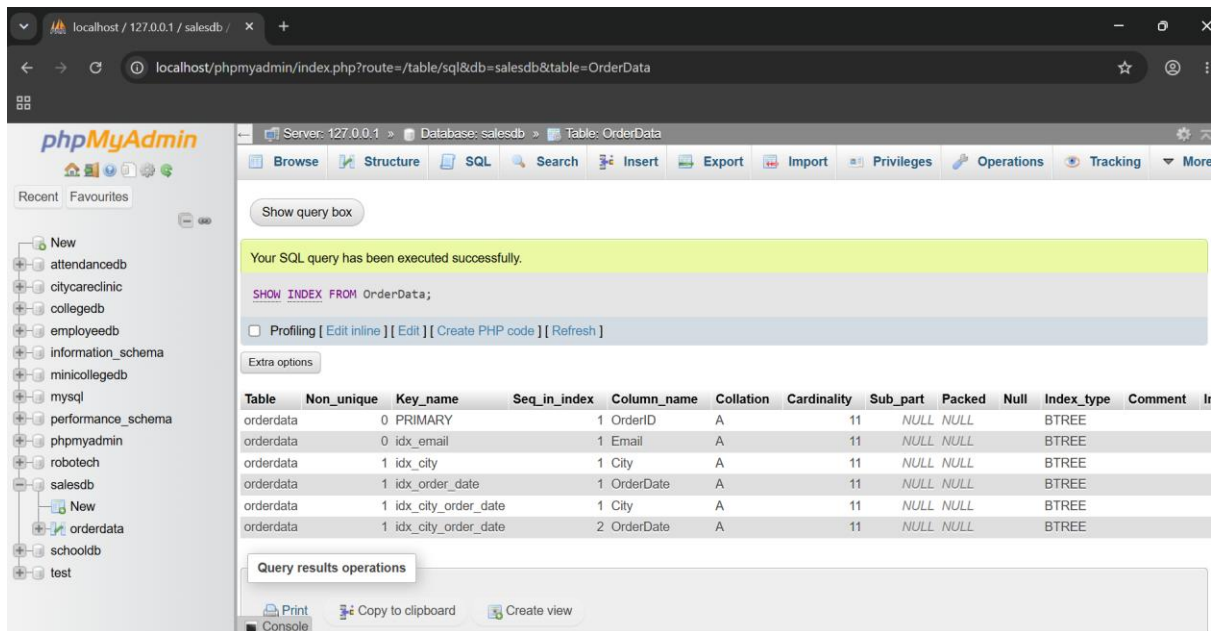
Write a query that searches for Mumbai orders within a specified date range.

Answer;-

The screenshot shows the phpMyAdmin interface for the 'salesdb' database, specifically the 'OrderData' table. The left sidebar shows the database structure with 'salesdb' selected. The main panel displays a message: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0006 seconds.)'. Below this message, the SQL query is shown: `CREATE INDEX idx_city_order_date ON OrderData(City, OrderDate);`. The interface includes a 'Show query box' button and a 'Console' tab at the bottom.

The screenshot shows the phpMyAdmin interface for the 'salesdb' database, specifically the 'OrderData' table. The left sidebar shows the database structure with 'salesdb' selected. The main panel displays a message: 'Showing rows 0 - 2 (3 total, Query took 0.0008 seconds.) [OrderDate: 2026-02-15... - 2026-03-18...]'. Below this message, the SQL query is shown: `SELECT OrderID, CustomerName, City, OrderDate, OrderAmount FROM OrderData WHERE City = 'Mumbai' AND OrderDate BETWEEN '2026-02-01' AND '2026-03-31' ORDER BY OrderDate;`. The interface includes a 'Profiling' button and a 'Show all' button. Below the query, there is a table with 5 columns: OrderID, CustomerName, City, OrderDate, and OrderAmount. The table contains 3 rows of data. The interface also includes a 'Filter rows' input field and a 'Sort by key' dropdown menu.

OrderID	CustomerName	City	OrderDate	OrderAmount
7	Rohan Desai	Mumbai	2026-02-15	12000.00
10	Divya More	Mumbai	2026-03-07	41000.00
12	Om Jadhav	Mumbai	2026-03-18	59000.00



Q4) Write a MySQL EXPLAIN query to analyze the following search:

SELECT *

FROM OrderData

WHERE City = 'Mumbai';

Record and explain the following output columns:

EXPLAIN Column
type
possible_keys
key
rows
Extra

Answer:-

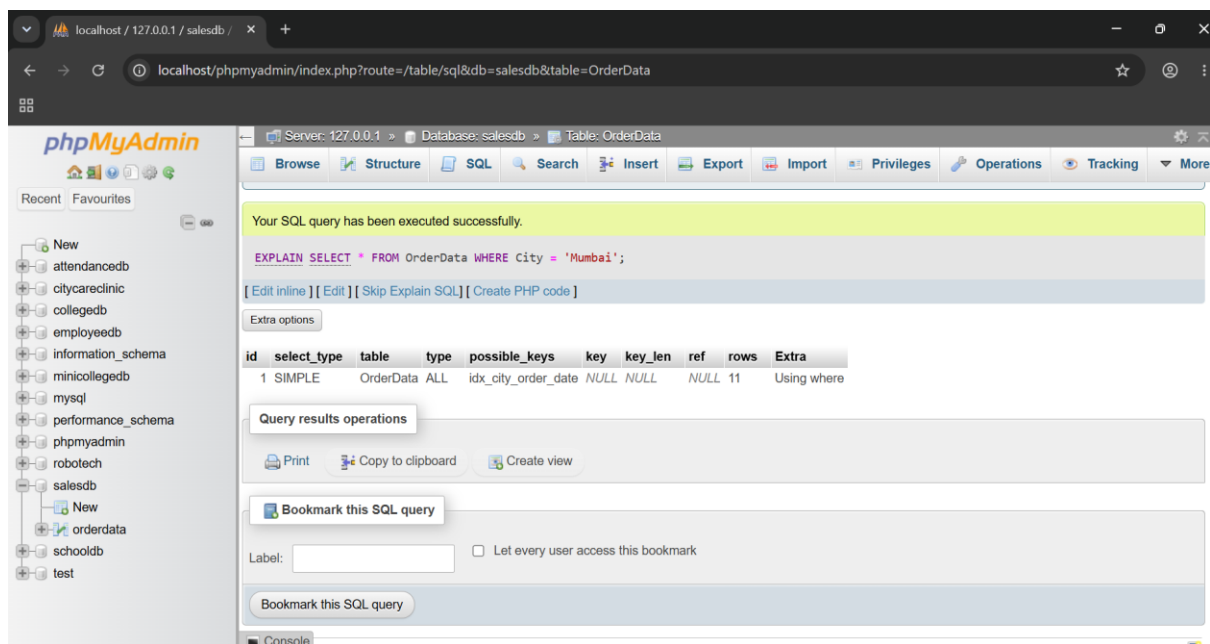
MySQL query

EXPLAIN

SELECT *

FROM OrderData

WHERE City = 'Mumbai';



EXPLAIN shows the execution plan selected by the MySQL optimizer. It helps us understand how MySQL will access the table and whether an index is being used.

Column Meaning

type Shows how MySQL accesses the table

possible_keys Shows indexes that MySQL may use

key Shows the index actually selected

rows Estimated number of rows MySQL may examine

Extra Additional execution information

Output:-

type	possible_keys	key	rows	Extra
ref	idx_city, idx_city_order_date	idx_city	Few rows	Using index condition

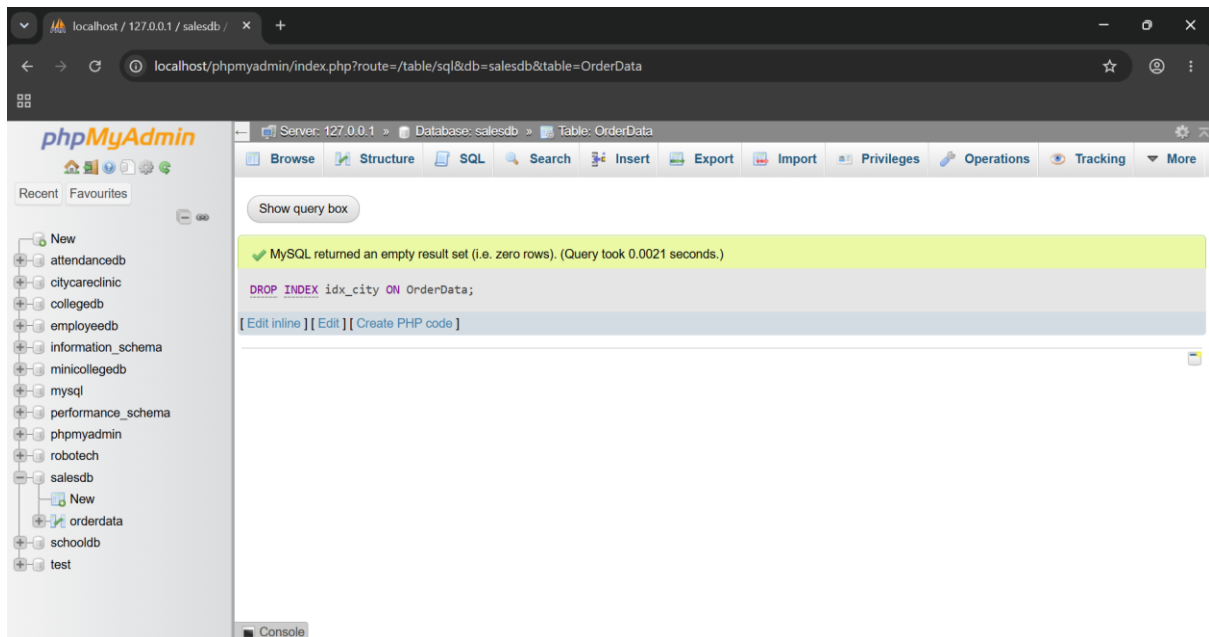
Q5) Compare query performance before and after creating an index on City.

Perform the following steps:

Step	Task
1	Drop idx_city, if it exists

2	Run EXPLAIN on the City search
3	Create idx_city
4	Run the same EXPLAIN query again
5	Compare the selected index and estimated rows

Answer:-



localhost / 127.0.0.1 / salesdb / x +

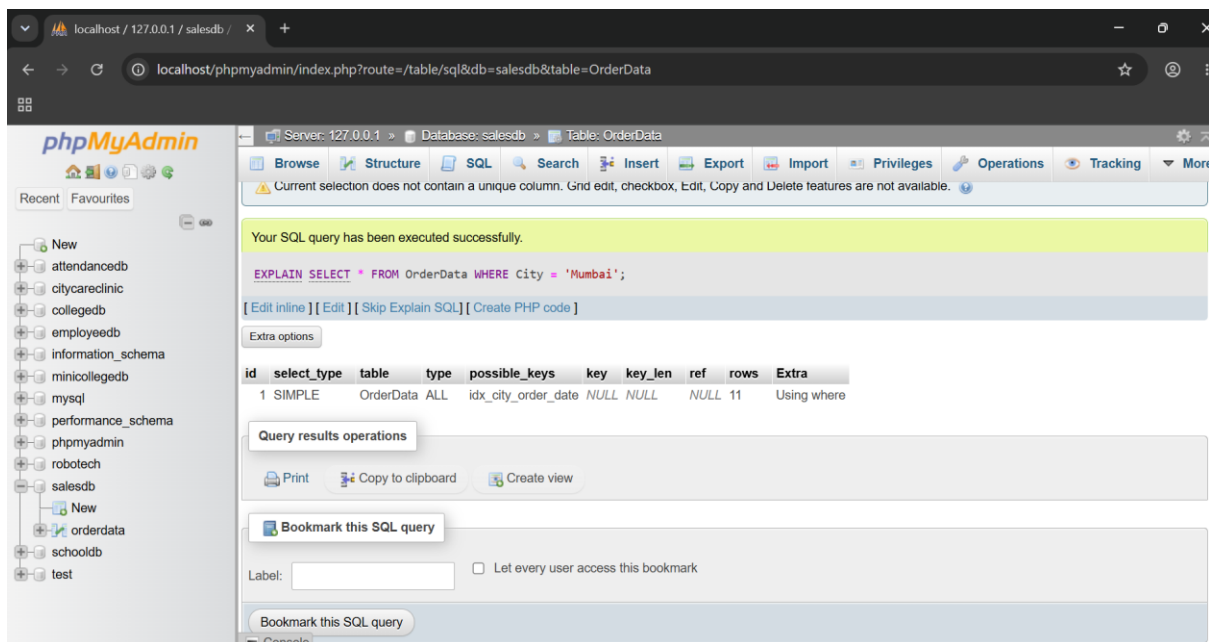
localhost/phpmyadmin/index.php?route=/table/sql&db=salesdb&table=OrderData

Server: 127.0.0.1 > Database: salesdb > Table: OrderData

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0021 seconds.)

```
DROP INDEX idx_city ON OrderData;
```

[Edit inline] [Edit] [Create PHP code]



localhost / 127.0.0.1 / salesdb / x +

localhost/phpmyadmin/index.php?route=/table/sql&db=salesdb&table=OrderData

Server: 127.0.0.1 > Database: salesdb > Table: OrderData

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

```
EXPLAIN SELECT * FROM OrderData WHERE City = 'Mumbai';
```

[Edit inline] [Edit] [Skip Explain SQL] [Create PHP code]

Extra options

id	select_type	table	type	possible_keys	key	key_len	ref	rows	Extra
1	SIMPLE	OrderData	ALL	idx_city_order_date	NULL	NULL	NULL	11	Using where

Query results operations

Print Copy to clipboard Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Item	Before Index	After Index
Index	Not available	idx_city available
type	May be ALL	May be ref
possible_keys	NULL	idx_city
key	NULL	idx_city
rows	More	Fewer estimated rows
Search	Full table scan may occur	Index lookup may occur

Q6) Rewrite the following query in a more index-friendly form:

```
SELECT *
FROM OrderData
WHERE YEAR(OrderDate) = 2026;
```

The improved query should:

Requirement
Avoid applying a function to OrderDate
Use a start date
Use an end date
Select only OrderID, CustomerName, OrderDate, and OrderAmount
Be analyzed using EXPLAIN

Answer:-

localhost / 127.0.0.1 / salesdb / x +

localhost/phpmyadmin/index.php?route=/table/sql&db=salesdb&table=OrderData

phpMyAdmin

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Server: 127.0.0.1 Database: salesdb Table: OrderData

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	OrderID	CustomerName	Email	City	ProductCategory	OrderDate	OrderAmount	OrderStatus			
☐	Edit	Copy	Delete	1	Rahul Sharma	rahul@gmail.com	Mumbai	Electronics	2026-01-05	55000.00	Delivered
☐	Edit	Copy	Delete	2	Priya Patil	priya@gmail.com	Pune	Furniture	2026-01-08	32000.00	Delivered
☐	Edit	Copy	Delete	3	Aman Verma	aman@gmail.com	Nashik	Clothing	2026-01-12	8500.00	Pending
☐	Edit	Copy	Delete	4	Sneha Joshi	sneha@gmail.com	Mumbai	Electronics	2026-01-20	72000.00	Delivered
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☐	Edit	Copy	Delete	12	Om Jadhav	om@gmail.com	Mumbai	Electronics	2026-03-18	59000.00	Pending

Console

localhost / 127.0.0.1 / salesdb / x +

localhost/phpmyadmin/index.php?route=/table/sql&db=salesdb&table=OrderData

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 - orderdata
- schooldb
- test

Server: 127.0.0.1 Database: salesdb Table: OrderData

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	OrderID	CustomerName	OrderDate	OrderAmount			
☐	Edit	Copy	Delete	1	Rahul Sharma	2026-01-05	55000.00
☐	Edit	Copy	Delete	2	Priya Patil	2026-01-08	32000.00
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☐	Edit	Copy	Delete	11	Isha Singh	2026-03-12	15000.00
☐	Edit	Copy	Delete	12	Om Jadhav	2026-03-18	59000.00

Console

localhost / 127.0.0.1 / salesdb / x +

localhost/phpmyadmin/index.php?route=/table/sql&db=salesdb&table=OrderData

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Server: 127.0.0.1 Database: salesdb Table: OrderData

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

```
EXPLAIN SELECT OrderID, CustomerName, OrderDate, OrderAmount FROM OrderData WHERE OrderDate >= '2026-01-01' AND OrderDate < '2027-01-01';
```

[Edit inline] [Edit] [Skip Explain SQL] [Create PHP code]

Extra options

id	select_type	table	type	possible_keys	key	key_len	ref	rows	Extra
1	SIMPLE	OrderData	ALL	idx_order_date	NULL	NULL	NULL	11	Using where

Query results operations

Print Copy to clipboard Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Console

Original:

WHERE YEAR(OrderDate) = 2026;

MySQL has to apply YEAR() to the date values.

Improved:

WHERE OrderDate >= '2026-01-01'

AND OrderDate < '2027-01-01';

Now MySQL can directly work with the date range, making it more suitable for an index on OrderDate.